



Storm-driven dispersal of fluvial fine sediment on a low-energy inner shelf

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ABSTRACT

Prolonged *in-situ* measurements and satellite images were analyzed to investigate storm-driven dispersal patterns of fluvially derived fine sediment in the Atchafalaya Bay/Shelf during winter storms and tropical cyclones and how they remotely influence fine-sediment distribution of an offshore sand shoal. It was revealed that storm-driven offshore advection coinciding with sediment re-suspension transports re-suspended sediment and fine sediment from the river and from the bay to the inner shelf regardless of the storm types. However, we found that the process leading to the dispersal varies with the types of storms, their intensity and tracks. On the shelf, while tropical cyclones were capable of bringing re-suspended sediment to the surface, winter storms limited the extent of sediment suspension up to approximately 10 m above the bottom. Thus, only the sediment suspension on the shallower part of the shelf (< 10 m in depth) was expected to contribute significantly to sediment dispersal during winter storms. The estimated mass of re-suspended sediment in the bay during a single storm event accounted for more than a quarter of the total sediment mass from the river in the entire summer; more than half of the re-suspended sediment was expected to be transported to the shelf. It was also found that the storm-driven advection plays an important role in the migration of the fine sediment plumes to the offshore shoal. Despite the patchy and ephemeral nature of fine sediment on the shoal, the findings have established an important link between storm-driven sediment dispersal and sediment dynamics of ecologically sensitive shoals along the Louisiana coast.

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1. Introduction

Fine sediment debouched from rivers is dispersed and eventually accumulated in a bay, estuary, and shelf seas and therefore, play a significant role in shaping the landscape of the nearshore and shelf environment. The sediment dispersal goes through at least four stages: (I) sediment supply, (II) initial deposition, (III) re-suspension and transport, and (IV) long-term net accumulation (Wright and Nittrouer, 1995). Many of the river systems in the world such as Mississippi, Amazon, and Chanjiang go through these stages. Patterns of the sediment dispersal are significantly controlled by morpho-hydrodynamic processes. Thus, understanding morpho-hydrodynamics and their implications for coastal and nearshore ecological processes is a key to more effective management of estuarine and coastal environment.

The Louisiana shelf, located in the northwest Gulf of Mexico, is a low-energy, micro-tidal shelf where sediment suspension is

largely limited to high-energy events (Adams et al., 1987; Kobashi et al., 2007; Pepper and Stone, 2004; Wright et al., 1997). While bottom sediment on the shelf consists primarily of silt and clay due to a large influx of fine sediment from two major rivers, there are several offshore sand shoals. The shoals have been recognized as biological hot spots and could serve as a refuge during summer hypoxic events for the benthic organisms, which favor sandy substrate (Stone et al., 2009).

What is unique about the shelf from a sedimentology perspective is that the high sediment load from two major rivers and variable winds associated with seasons, quasi-periodic passage of winter storms, and occasional landfall of tropical cyclones are thought to create complex sedimentary environment and potentially influence benthic habitats on the shelf. For example, it was reported that one of the offshore sand shoals, Ship Shoal, located 65 km southeast of the mouth of the Atchafalaya River (see Fig. 1 for location), experienced intermittent exposure to fine sediment (silt and clay) (Stone et al., 2009). Though it is not yet clear on the sediment sources and the morpho-hydrodynamic processes that facilitated this changing sediment textures of the shoal, we have hypothesized based on semi-idealized advection–dispersion

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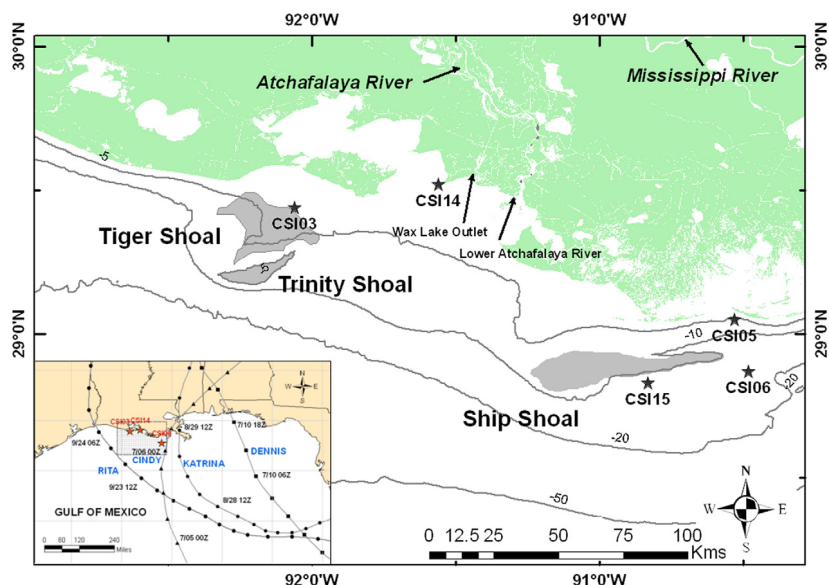


Fig. 1. Map of the study area. Gray-shaded areas show offshore sand shoals. Star marks are WAVCIS stations.

modeling that storm-driven sediment dispersal contributes at least partially to the deposition of fine sediment on the shoal (Kobashi, 2009). The potential link between sediment dispersal and the delicate benthic ecology of the shoal highlights an importance of the subject to be investigated.

In the past, the studies of sediment dispersal of the shelf off the Atchafalaya River focused largely on specific aspects of the sediment dispersal for different dispersal stages: structure of sediment plume (Wells and Kemp, 1981), response of sediment plume to storm winds based largely on satellite image analysis (Walker, 2001; Walker and Hammack, 2000), storm and flood deposits based on short-term field measurements and short-lived radioisotope geochronology (Allison et al., 2000, 2005), and sediment deposition and advection/diffusion based on numerical modeling (Kobashi, 2009; Xu et al., 2011). These studies have improved our understanding of various aspects of sediment dispersal of the region; however, it is not well understood the role of sediment suspension and advection associated with variable storm winds on sediment dispersal in particular during low-discharge periods.

In this paper, we discuss storm-driven sediment dispersal with a focus on suspension and transport and its potential influence on the sedimentary environment of Ship Shoal based on prolonged (month-long and beyond) in-situ data from multiple field stations that have been deployed in the region and satellite images from MODIS (Moderate Resolution Imaging Spectroradiometer) satellites (Terra and Aqua). The focus is on two contrasting storm conditions, namely, winter storms and tropical cyclones that made landfall in the region between October 2004 and September 2005.

2. Winter storms and tropical storms

Winter storms, a low-pressure system originating in the arctic polar region (polar frontal zone), accompany atmospheric fronts (i.e. cold fronts and warm fronts). They move from west to east following the prevailing westerly wind; the fronts eventually pass the northern Gulf coasts every 3 to 10 days between October and May (DiMego et al., 1976; Hsu, 1988). The passage of a front accompanies a sudden change in wind direction, temperature and barometric pressure. The strong wind fields and the rapidly fluctuating wind direction are known to affect short-term hydrodynamics and sediment transport along the low-energy, micro-tidal

Table 1
Summary of tropical storm and hurricanes that made landfall in the 2005 hurricane season.

Name	Maximum wind speed	Minimum pressure	Time of landfall	Landfall location
TS. Arlene	31 m/s	989 mb	19:00 UTC, June 11th	Pensacola, FL
H. Cindy	34 m/s	956 mb	03:00 UTC, July 6th	Grand Isles, LA
H. Dennis	64 m/s	946 mb	19:30 UTC, July 10th	Santa Rosa Islands, FL
H. Katrina	73 m/s	902 mb	11:10 UTC, August 29th	Buras, LA
H. Rita	78 m/s	897 mb	07:40 UTC, September 24th	Johnson's Bayou, LA

shelf (e.g., Kobashi et al., 2007; Pepper and Stone, 2004; Stone, 2000). Between October 2004 and May 2005, approximately 40 cold fronts passed the Louisiana Coast.

As for tropical cyclones, 27 named tropical storms were spawned in the Atlantic Basin during the 2005 hurricane season, out of which 15 were developed into the hurricane strength (i.e. maximum sustained wind speed higher than 119 km/hour or 74 mph). The season is the most active in the recorded history (National Hurricane Center (NHC), 2006). For the five tropical storms and hurricanes that hit the Gulf coasts in 2005, Arlene, Cindy, Dennis, and Katrina made landfall east of our study area and Rita made landfall west of the study area. All hurricanes caused considerable damage along the coastal Louisiana due to their hurricane-forced winds, waves and high storm surge. Even though Hurricane Dennis made landfall in the Florida Panhandle, further east of the study area, hurricane-induced winds and waves caused a significant damage to the barrier islands along the eastern Louisiana.

Table 1 shows a brief summary of the tropical storm and hurricanes, which made landfall in the northern Gulf of Mexico in 2005. More detailed information of Hurricanes Katrina and Rita is provided by Knabb et al. (2005a,b).

3. Methods

The data sets used for analysis include *in-situ* data from near real-time ocean observing stations, WAVCIS (Wave-Current-Surge

Information System. www.wavcis.lsu.edu) CSI's-3, 6, and 14 (see the locations in Fig. 1) (Stone et al., 2001), and satellite images from the Louisiana State University's Earth Scan Laboratory (hereafter referred to as ESL. www.esl.lsu.edu). In addition, streamflow at Simmesport (station#03045Q), Louisiana (~180 km from the Atchafalaya River mouth) was obtained from US Army Corps of Engineer (USACE) and streamflow, sediment load, and suspended sediment concentration (hereafter SSC) at Morgan City (station #07381600) (~45 km from the river mouth) and Wax Lake Outlet (station#07381590) (~24 km from the river mouth) were both obtained from the US Geological Survey's National Water Information System (USGS NWIS) (Baumann et al., 2006). USACE provides daily streamflow data and the USGS NWIS provides streamflow, sediment load, and SSC data with the frequency of approximately once a month. WAVCIS CSI-14 located in the Atchafalaya Bay provides hourly directional wave parameters, single-point current velocity, temperature, salinity, and turbidity data at 1.6 m above the bottom as well as meteorological data. CSI-3 and CSI-6, both located on the shelf (approximately 5 m and 20 m in depth, respectively), provide hourly wave parameters, water level, and current profiles. The detailed information regarding WAVCIS is provided by Stone et al. (2001).

From the meteorological data, wind stress (τ_{wind}) was calculated using the following bulk equation.

$$\tau_{wind} = \rho_a C_f U_{10}^2 \quad (1)$$

where ρ_a is air density (1.3 kg m^{-3}), C_f is surface drag coefficient, and U_{10} is wind speed at 10 m above the surface. The drag coefficient (C_f) was computed based on Holthuijsen (2007) and Wu (1982). SSC at CSI-14 (deployed at West Cote Blanche Bay, see Fig. 1) was estimated by calibrating the turbidity sensor (optical backscatter sensor, hereafter referred to as OBS), using sediment and water samples collected from the nearby site. The detailed calibration procedure is given in Sheremet et al. (2005). The conversion from the OBS to SSC was obtained using the following formula ($r^2 = 0.998$) (Liu, B. Personal communication).

$$SSC = 0.001 * OBS - 0.0114 \quad (2)$$

For CSI's-3 and 6, we used acoustic backscatter signal intensity (hereafter referred to as ABS) from ADCP (Acoustic Doppler Current Profiler) as a proxy for the SSC profile (SonTek®, 1997).

Except for a few isolated transgressive sand bodies, the bottom sediment of the study area is mainly silt and clay. The routines for computing the bottom boundary layer parameters (hereafter referred to as BBLP) used in previous studies are strictly applicable for the sandy bottom (cf. Nielsen, 1992; Pepper and Stone, 2004). For cohesive bed, it is known that the numerical algorithms to compute the BBLP are different from those for sands (e.g. McAnally et al., 2007; Mehta, 1989). Moreover, bottom drag depends on sediment concentration and can be reduced by up to 50% depending on the magnitude of the concentration. Hence, the BBLP using conventional methods such as the quadratic stress law may be overestimated (cf. Li and Gust, 2000).

Unfortunately, as with often the case with many cohesive sediment studies using in-situ data, not all data sets necessary to estimate the BBLP for cohesive sediment were available for this study. Thus, this study does not discuss some aspects of cohesive sediment dynamics such as settling and flocculation. As mentioned previously, the focus of this paper is fine sediment suspension and transport. We attempted to discuss them using the best available information as mentioned below. It is also noted that wave-current interactions are not discussed in this paper because in the study area, waves are the primary force to induce suspension and current to transport suspended sediment (Kobashi, 2009). Treating wave and current separately suits the objective. The method was used in a previous study discussing

fine sediment dynamics on a sandy shoal (Kobashi and Jose, 2018).

The bottom shear stress due to waves (τ_w) was estimated based on the following equation (Whitehouse et al., 2000).

$$\tau_w = \frac{1}{2} \rho_f f_w u_{ob}^2 \quad (3)$$

where ρ_f is seawater density, f_w is wave friction factor, and u_{ob} is near-bottom wave orbital velocity. The friction factor for smooth bed was estimated as follow.

$$f_w = \begin{cases} 2.0 \cdot R_w^{-0.5} & (R_w \leq 5 \times 10^5) \\ 0.0521 \cdot R_w^{-0.187} & (R_w > 5 \times 10^5) \end{cases} \quad (4)$$

R_w is wave Reynolds number ($= (u_{ob} \cdot A) / \nu$, where A is semi-orbital excursion and ν is kinematic viscosity) (Whitehouse et al., 2000).

The bottom shear stress due to current (τ_c) was calculated by the quadratic stress law (Thompson et al., 2006).

$$\tau_c = \rho_f C_{D100} \bar{u}_{100}^2 \quad (5)$$

where C_{D100} is drag coefficient, and $\bar{u}_{100}$ is mean current at 1 m above the bottom. Wave shear stress was computed using the data from CSI's-3, 6, and 14, and current shear stress was calculated using the CSI-14 data. Since the height of the pressure and current sensors at CSI-14 was 1.6 m above the seabed, we extrapolated the current to that of the conventional height of 1 m above the bottom, by the logarithmic method, assuming that flow at the bottom was zero. For all current data, high-frequency oscillations of currents such as tidal and inertial currents were removed by filtering out signals shorter than 40 h since our focus is on storm-induced currents. For the bottom drag coefficient, Thompson et al. (2006) presented a relationship between the drag coefficient and sediment concentration based on lab experiment. The drag reduction occurred in their experiment for the sediment concentration between 5 and 20 g l^{-1} . For the concentration lower than the range ($0\text{--}5 \text{ g l}^{-1}$), the drag coefficient for turbid water was close to the one for clear water. The sediment concentration measured for the study area was mostly lower than 1 g l^{-1} ($=1 \text{ kg m}^{-3}$). Sheremet et al. (2005) presented a similar value of SSC at 1 m above the bottom for CSI-3 station during Hurricane Claudette. However, the result of a 1-D numerical model they implemented suggested a lutocline near the bottom with the concentration higher than 10 g l^{-1} during the waning phase of the storm, for which the drag reduction likely comes into play. This is in line with the near-bottom SSC higher than 20 g l^{-1} reported by Allison et al. (2000) off the Atchafalaya Bay. Though the above facts suggest the reduction in the drag coefficient, the effect of the drag reduction was not included in this study and a constant value of 0.0022 for C_{D100} , following Soulsby (1997) for muddy bottom, was used to estimate the current shear stress. Consequently, the shear stress due to current may be overestimated. The overestimation associated with the drag reduction is considered not to significantly influence the discussion of this paper because the current shear stress played a minor role in sediment suspension for the region as mentioned in the following sections. In the past, similar values of the drag coefficient were used to study sediment dynamics on muddy bottoms (e.g., Chuang and Wiseman, 1983; Adams et al., 1987). Despite such simplified treatments of cohesive sediment dynamics, we were able to discuss salient features of morpho-hydrodynamics and fine sediment dispersal associated with storms on the Atchafalaya Bay and the adjoining inner shelf.

The critical shear stress for suspension of 0.15 N m^{-2} was used as the threshold for cohesive sediment suspension with reference to Wright et al. (1997) for their study in the eastern Louisiana inner shelf. The value was empirically derived and the same value has also been used for other cohesive sediment studies (Dale Kerper, personal communication).

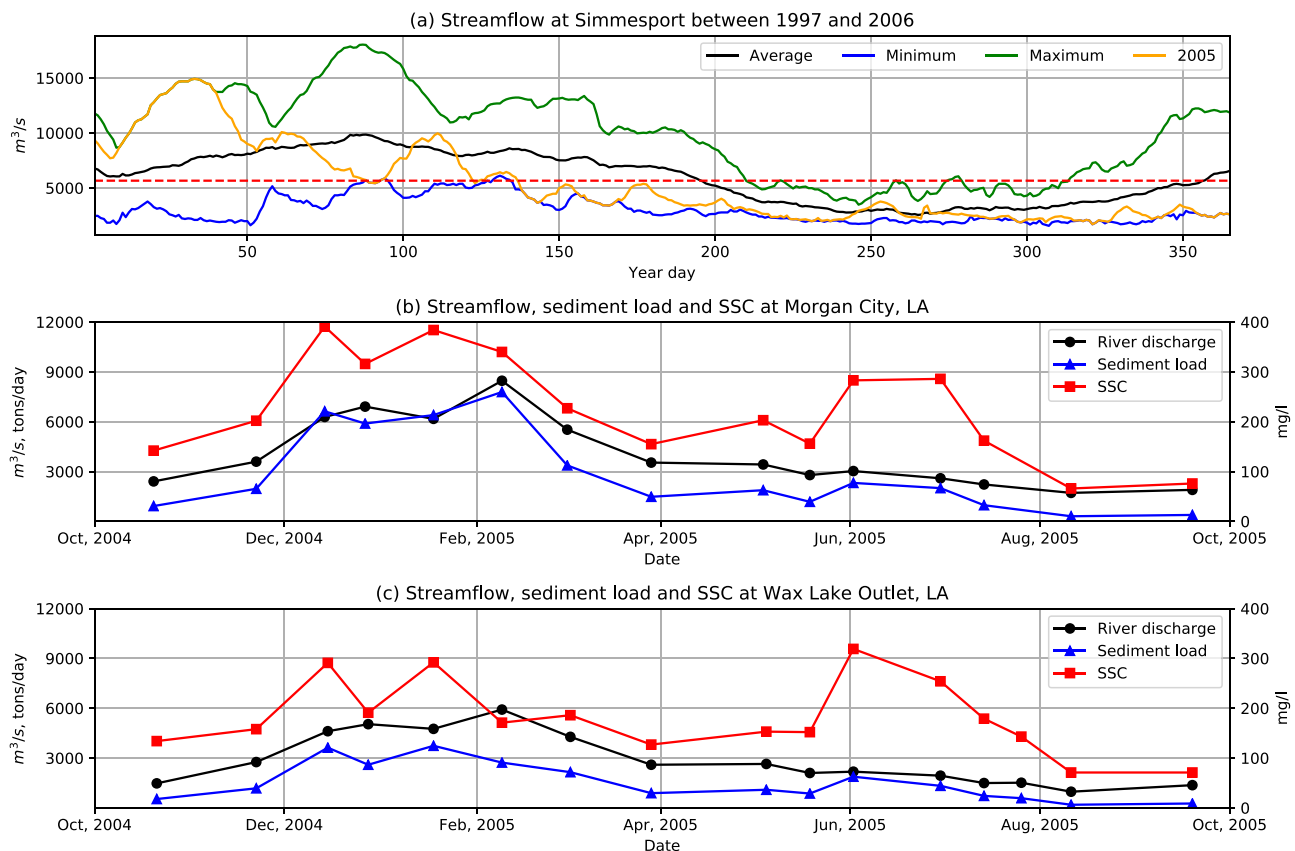


Fig. 2. (a) Streamflow at Simmesport, Louisiana for the period between 1997 and 2006. The dash line shows the threshold of high river discharge ($5662 \text{ m}^3/\text{s}$), (b) streamflow, sediment load and SSC at Morgan city, LA, and (c) streamflow, sediment load, and SSC at Wax Lake Outlet.

4. Results

4.1. Atchafalaya River freshwater discharge and sediment load

In Fig. 2a, the maximum, minimum and average river discharge at Simmesport, located upstream of the Atchafalaya River, for the period between 1997 and 2006 is presented. The discharge is maximum in spring (March) and minimum in summer (September). In 2005, the discharge was highest at the beginning of the year, which was higher than $5662 \text{ m}^3 \text{ s}^{-1}$, the value suggested by Walker and Hammack (2000) as a high-discharge event. However, the discharge was below average for most of the year (Fig. 2a). Freshwater discharge, sediment load, and SSC at Morgan City (upstream of the lower Atchafalaya River mouth) and Wax Lake Outlet are shown in Fig. 2b and c. Note that the Atchafalaya River is bifurcated into two distributaries: the lower Atchafalaya River and Wax Lake Outlet (See the location of both rivers in Fig. 1). The sediment load was well correlated with the freshwater discharge for both stations. The freshwater discharge from Morgan City and Wax Lake Outlet reached $14,385 \text{ m}^3 \text{ s}^{-1}$ in February and the corresponding sediment load was estimated to be $0.37 \text{ million tons day}^{-1}$ (see Fig. 2b and c). From Fig. 2b and c, the total annual sediment load estimated for the Lower Atchafalaya River and Wax Lake Outlet was 32.4 and 17.8 million tons, respectively. One of the noticeable characteristics of the data for both Morgan City and Wax Lake Outlet is that the SSC was abnormally high during summer (June–July), in spite of low-river discharge, low sediment load, and low wind energy regime. There was no correlation between the two data sets and river discharge in Fig. 2; there were no high-energy events during the period. The above facts suggest that the possible reasons for the high SSC could have been upstream dredging, navigation activities, land clearing and/or agricultural practices.

4.2. Wind variability associated with the storms

Wind regime during winter storms and tropical cyclones varied dramatically in a short period of time (2–3 days; Fig. 3). Three representative cases are presented and illustrated in Fig. 3. For a winter storm, pre-frontal easterly/southeasterly winds quickly shifted to westerly/north-westerly winds when the storm passed over the region (Fig. 3 left column). The change in wind direction was associated with the passage of a cold front. The wind during winter storms is, therefore, characterized by the change in direction and strong post-frontal winds. For Hurricane Katrina, which made landfall east of the study area, progressive changes in wind direction were similar to those of the winter storm except for the strong offshore wind when the storm came closer to the land (Fig. 3 middle column). For Hurricane Rita, which made landfall west of the study area, the easterly wind prevailed. In addition, an offshore component of the wind prevailed when Rita was located further offshore and an inshore component of the wind prevailed while the storm approached the land (Fig. 3 right column). As seen from satellite images and presented in the following subsections, we found that those variable wind fields play a role in sediment dispersal patterns.

4.3. Distinct fine sediment dispersal patterns from satellite images

Fig. 4 presents a series of MODIS satellite images obtained from the ESL. The westward transport of fine sediment is evident during fair weather (Fig. 4a). Following the passage of a cold front, satellite image captured a dispersal shift to the south/southeast in line with Walker and Hammack (2000) (Fig. 4b). During hurricane landfalls (Fig. 4c for Hurricane Katrina, Fig. 4d for Hurricane Rita), the satellite images indicated that the sediment was spread

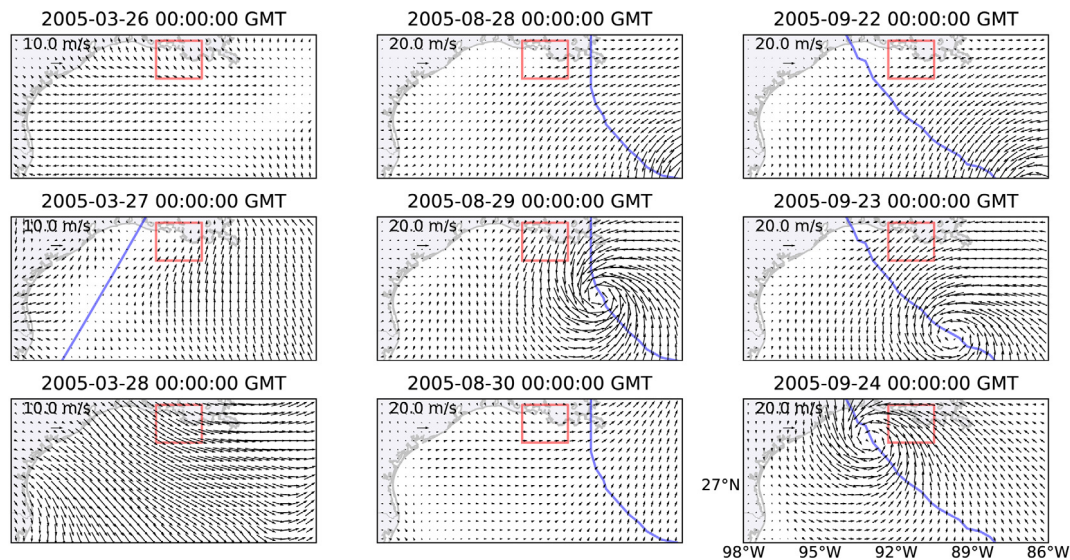


Fig. 3. Wind vectors during a typical winter storm (left column), Hurricane Katrina (middle column) and Hurricane Rita (right column). The red squares are the location of our study area. The blue lines represent a cold front and hurricane tracks.

broadly over the shelf, in particular in Fig. 4d, despite low river discharge during summer. It is noted that the satellite image in Fig. 4c was taken several days after the landfall. Though the image suffered cloud coverage, Fig. 4c showed a similar pattern that can be seen in Fig. 4d. During fair weather, fine sediment plume did not reach up to the Ship Shoal area (Fig. 4a); whereas, during winter storms and tropical cyclones, it can be seen that a blanket of fine sediment covered the offshore shoals, not just Ship Shoal, but also Tiger and Trinity shoals (see the location of the shoals in Fig. 1). The satellite images in Fig. 4 suggest that sediment dispersal can be significantly influenced by storm-driven currents and sediment suspension.

4.4. Salient features of ocean circulation and sediment dynamics on the bay/shelf

Fig. 5 presents a time series of wind stress, wave height, mean currents and ABS at CSI's-3, 6, and 14. Variability of waves, currents, bottom boundary layer parameters and SSC over the Atchafalaya bay/shelf was strongly associated with meteorological conditions and river discharge from the Atchafalaya River. When storms passed over the study region (reversed triangles with dash lines on Fig. 5a), increases in wave height, current speed, bottom shear stress due to waves, and SSC are evident (Fig. 5). The bottom shear stress that exceeds the threshold for sediment suspension (i.e. critical shear stress. Purple dashed line in Fig. 5e) coincided with the elevated ABS at CSI's-3 and 6 and it largely occurred during the passage of cold fronts and tropical cyclones (Fig. 5e, f, g). The southward (offshore) current at CSI-14 corresponded to post-frontal winds and the current was quickly reversed after the front passed (Fig. 5a, d). These short-term oscillations, with a period of 5 to 15 days, for current fields in winter and spring, corresponded to sudden changes in winds as a result of the passage of a cold front (Fig. 5c,d) and are a common feature along the northern Gulf coasts. For alongshore current, the westward current prevailed with occasional reversals associated with storms (Fig. 5d). River plume is transported downcoast (i.e. westward for our study region) (e.g. Wells and Kemp, 1981; Yankovsky and Chapman, 1997) and the prevailing westward current at CSI-14 located in the vicinity of a river mouth is thought to have shown this downcoast transport. The structure of river plume is also known to be influenced by winds (e.g. O'Donnel,

2010) and the transport is thought to have been enhanced by the southeasterly wind prevailing in the northwest Gulf of Mexico. The westward current weakened in summer and its direction was occasionally reversed, which was largely consistent with the wind reversal from downcoast to up-coast during summer months (Cochrane and Kelly, 1986).

At CSI-6, the wave height reached 2 m during winter storms and exceeded 4 m during tropical cyclones in 20 m of water (Fig. 5b). Wave height decreased as waves propagate inshore. The reduction in the wave height is considered to be the combination of wave dissipation due to bottom friction and also wave damping over the muddy seabed as reported by Sheremet and Stone (2003) for the region. Waves in the form of swells propagate primarily from southwest following the prevailing wind (Fig. 5b); however, when storms pass the region, the fetch-limited waves directing southeastward due to post-frontal and hurricane winds are generated and the waves become bi-modal or even multi-modal (Stone et al., 2005). Thus, the wave height should have been the result of the modal waves (Jose et al., 2007).

The high wave energy during storms led to sediment suspension at the bottom. High ABS, thus high SSC, corresponded to the elevated wave height and consequent wave-induced sediment re-suspension as explained in the following sub-section. Such variability is noticeable at CSIs 3 and 6, which are located on the shelf (5 m and 20 m in depth, respectively). Sediment suspension was prevalent during storms. For the deeper part of the shelf (e.g. CSI-6), the suspension was limited to 5–10 m above the bottom during winter storms; while, high SSC was noticed throughout the water column during hurricanes due to strong sediment re-suspension and storm-induced turbulence (vertical velocity of 10–15 cm s⁻¹ was recorded by ADCP) (Fig. 5g). For the shallower part of the shelf (e.g. CSI-3 with ~5 m in depth), the re-suspended sediment was often mixed throughout the water column regardless of the types of storms (Fig. 5f). During fair weather, on the other hand, wave-induced bottom shear stress was generally lower than the threshold for sediment suspension and the corresponding increase in SSC was not visible (Fig. 5e). The result is supported by the satellite images presented in Fig. 3.

4.5. Response of the sediment dispersal to two contrasting storms

The response of sediment dispersal to two contrasting storms was investigated using *in-situ* ocean observing data from WAVCIS

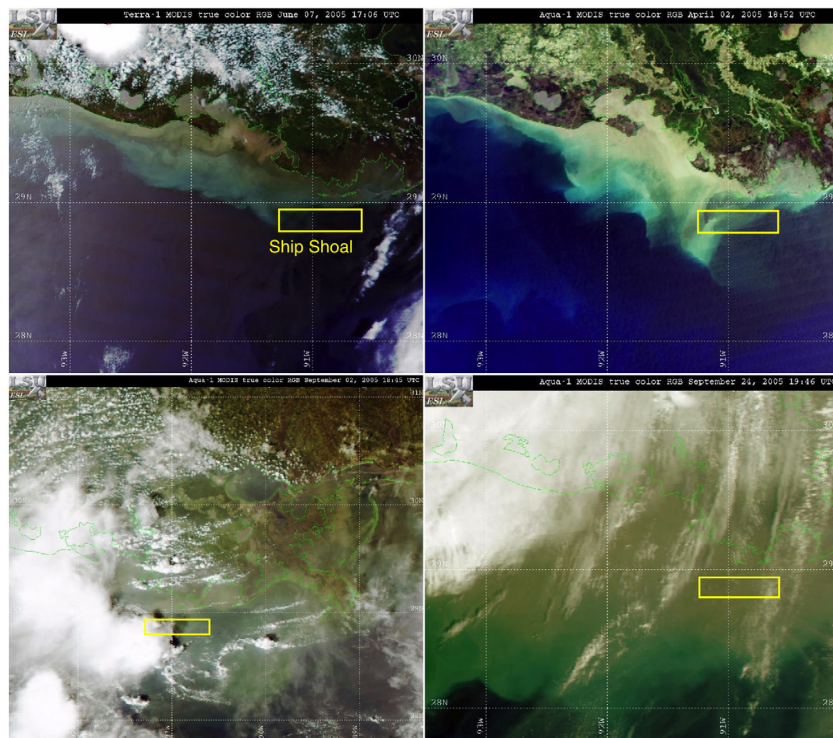


Fig. 4. MODIS satellite images taken over the Atchafalaya bay/shelf for different dates: (top left) June 7, 2005, (top right) April 2, 2005, (bottom left) September 2, 2005, and (bottom right) September 24, 2005. Yellow rectangles show the location of Ship Shoal.

CSI-14 (Figs. 6–8). Two hurricanes along with winter storms were selected: Hurricanes Katrina and Rita, which made landfall on the east and west of our study area, respectively.

In Fig. 6, time series data from CSI-14, for the time period between March 25th and April 6th, 2005, is presented. During this period, two winter storms passed through the study area: one on March 27th and the other on April 1st. During these events, the wind speed exceeded 10 m s^{-1} and the direction of wind strongly veered clock-wise from the southeast to from north (Fig. 6a, b). Wave height reached 0.5 m in depth of approximately 3 m on March 28th and simultaneously the bottom shear stress due to waves (τ_w) exceeded the threshold for sediment suspension (purple dashed line in Fig. 6d) consistent with an increase in SSC. Whereas, the shear stress due to currents (τ_c) was less than the threshold and was not correlated with the SSC during this period. Maximum shear stress was close to 0.4 N m^{-2} and maximum SSC was slightly higher than 0.4 kg m^{-3} during the post-frontal phase of the winter storm. The increase in SSC coincided with the southeasterly current generated by strong post-frontal winds.

Time series data from CSI-14 during Hurricane Katrina is presented in Fig. 7. Hurricane Katrina made landfall at approximately 200 km east of the study area, on August 29th, 2005. Wind speed reached 20 m s^{-1} and wave height exceeded 0.5 m at CSI-14, in spite of the fact that waves are fetch-limited in the bay due to the northerly wind (the station is located at the front left quadrant of the hurricane path). Shear stress due to waves reached 0.5 N m^{-2} during this time, which is more than three times higher than the estimated threshold for sediment suspension within the bay. The computed stress due to currents was poorly correlated and was always lower than the threshold. The SSC recorded the maximum value of 1.1 kg m^{-3} after the landfall and was associated with high wave-induced shear stress, clearly indicating strong sediment re-suspension and mixing during the post-hurricane phase. Both alongshore ($\sim 0.8 \text{ N m}^{-2}$) and cross-shore wind stress ($\sim 1.0 \text{ N m}^{-2}$) during the hurricane landfall was more than 40 times higher than those during fair weather conditions ($\sim 0.02 \text{ N m}^{-2}$ for both

alongshore and cross-shore wind stress). The northwest wind generated southeasterly current during the post-landfall phase. This offshore current coincided with high SSC contributed to offshore transport of fine sediment. Along with strong sediment suspension on the inner shelf (Fig. 5), the result validated the dispersal patterns identified from the satellite image obtained after the landfall (Fig. 4 bottom left).

In Fig. 8, time series data from CSI-14 during Hurricane Rita is presented. The hurricane made landfall at approximately 250 km west of the study area, near the Louisiana-Texas border on September 24th, 2005 (Fig. 3). The data showed that the maximum sustained wind speed at CSI-14 reached 24.4 m s^{-1} and wave height attained 1.8 m, approximately 18 times higher than the height during fair weather ($\sim 0.1 \text{ m}$), due to wave setup from the southerly hurricane-forced wind regime (cf. Fig. 3). The high waves induced strong sediment re-suspension, with the shear stress due to waves reaching 0.75 N m^{-2} , which is five times higher than the threshold. While, the stress due to currents was still less than the threshold and had a poor correlation with the SSC. The SSC reached 1.45 kg m^{-3} and persisted for about a day. During the pre-hurricane landfall, wind direction progressively changed from the east, southeast, and south as the hurricane approached the land. During the approaching phase of the storm, the northwesterly current transported the re-suspended fine sediment towards onshore across most of the Atchafalaya bay and shelf (Figs. 5 and 8); while, during the post-landfall phase, the southerly return flow, transported the sediment back offshore although the wind direction was still from south (onshore wind) (Fig. 8). Freeman et al. (2015) also reported this return flow from the Sister Lake, Louisiana during the landfall of Hurricane Rita.

4.6. Frequency of dispersal shift and its impact on a sand shoal

As supported by the satellite images presented in Fig. 3, fine sediment dispersal episodes influence bottom sediment textures of Ship Shoal during both types of storms: tropical cyclones and

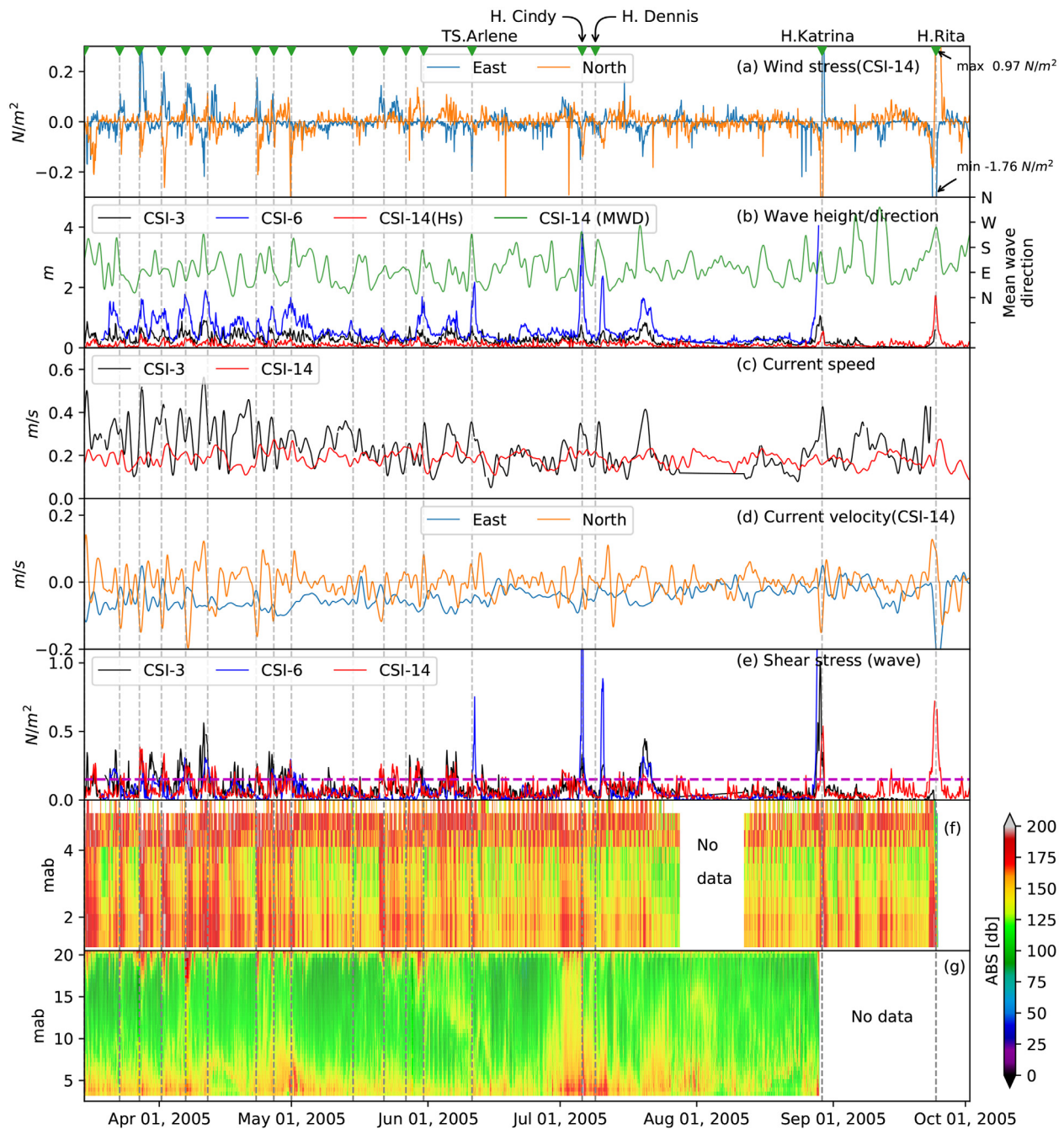


Fig. 5. A time series of (a) wind stress, (b) wave heights and direction, (c) low-passed current speed, (d) low-passed current velocities at CSI-14, (e) shear stress due to waves (τ_w), (f) ABS at CSI-3 and (g) ABS at CSI-6. Green reversed triangles on (a) show the time when winter storms passed or hurricanes made landfalls over the study region.

winter storms. Based on *in-situ* met-ocean data from WAVCIS CSI-3 and satellite images from the ESL between October 2004 and May 2005, we further analyzed the frequency of dispersal shifts that occurred during the study period. Our focus is on winter storms, not tropical cyclones since the quasi-periodic passages of cold fronts lead to more frequent exposure of fine sediment to the shoal than tropical cyclones. We defined here the dispersal shift as the persistent reversal of wind and coastal current direction from west/northwest to south/southeast during post-frontal phase of the storm. The number of plume reversals that reached Ship Shoal was determined based solely on satellite images from the ESL. Such satellite image analysis strongly depends on cloud coverage. Fortunately, cloud-less images were available for a few

days after each front passed (e.g. Fig. 4b). Thus, we were able to estimate the frequency of fine sediment that reached the shoal. Fig. 9 shows that the frequency of the dispersal shifts that reached the maximum during spring with a total of five shifting events in a month. The average number during an eight-month period was 0.125 times per day (i.e. once every 8 days). However, it was found that not all dispersal shifts resulted in the migration of the plumes far south into the shoal area; rather the sediment plumes transported by the reversed current appear to have reached the shoal once every 19 days on average. The number of the plume shifts had no correlation with the monthly mean Atchafalaya river discharge data (Fig. 9, square marks). However, it had a positive correlation with the monthly mean wind stress (Fig. 9,

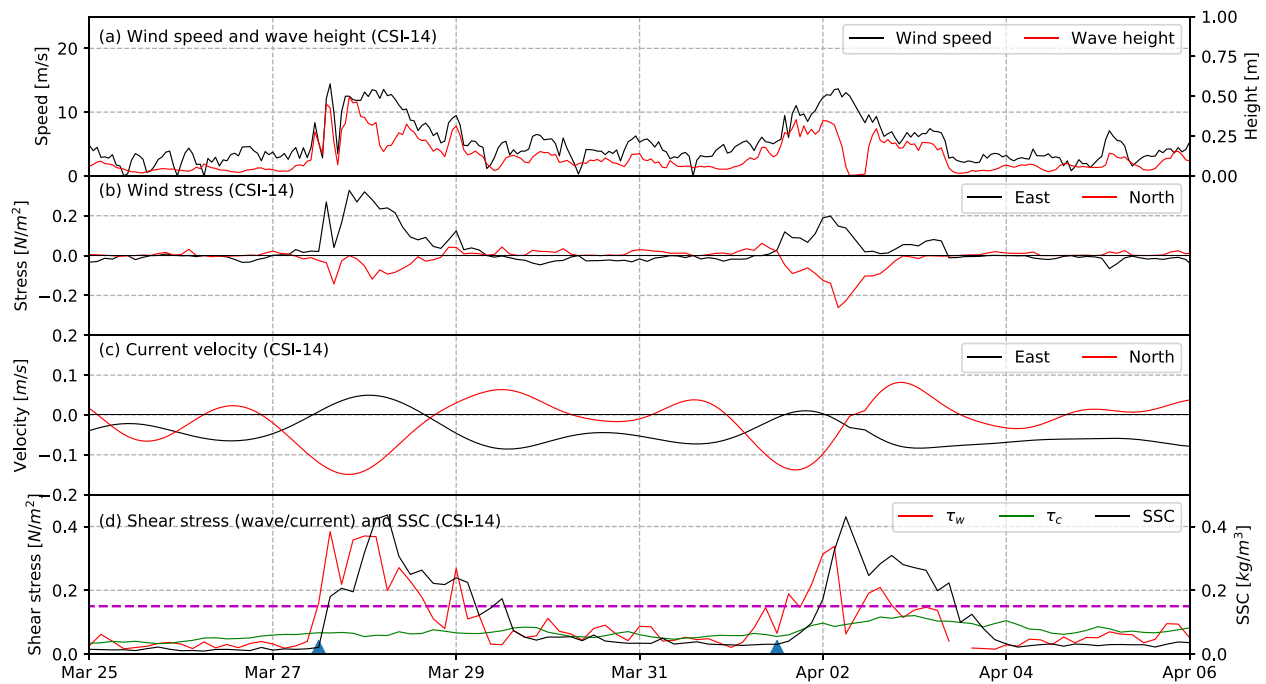


Fig. 6. A time series of (a) wind speed and wave height, (b) wind stress, (c) low-passed current velocities, (d) shear stress due to waves (τ_w) and current (τ_c) as well as SSC at CSI-14 for the period between March 25 and April 6, 2005. Purple dashed line in (d) represents threshold for sediment suspension (i.e. critical shear stress). Blue triangles indicate the time when cold fronts passed over the study region.

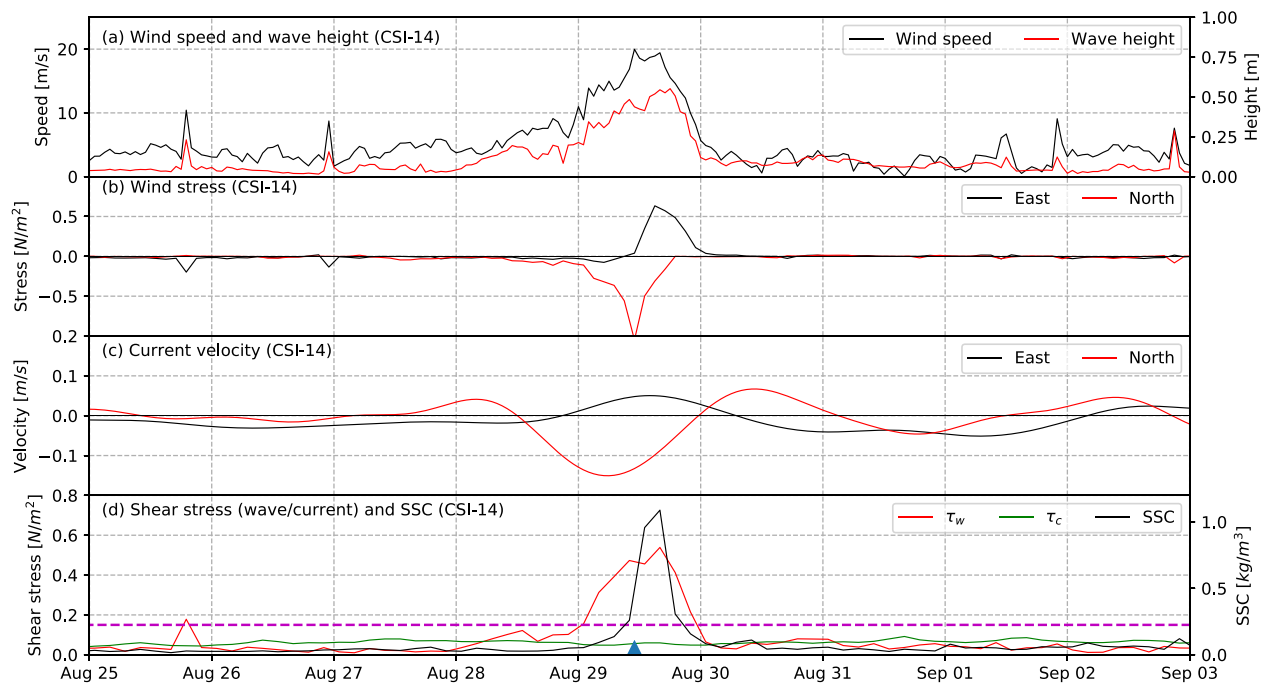


Fig. 7. A time series of (a) wind speed and wave height, (b) wind stress, (c) low-passed current velocities, (d) shear stress due to waves (τ_w) and current (τ_c) as well as SSC at CSI-14 for the period between August 25 and September 3, 2005. Purple dashed line in (d) represents threshold for sediment suspension (i.e. critical shear stress). Blue triangle indicates the time when Hurricane Katrina made landfall.

cross marks). Therefore, it is suggested that strong post-frontal wind stress may be a primary contributor to the distribution of fine sediment on the shoal. The result is in line with the data presented in this paper.

5. Discussion

In this study, we investigated how variable winds influence morpho-hydrodynamics and the resultant dispersal patterns of fluvially derived fine sediment in a bay and the adjoined shelf. We further looked into how the dispersal patterns potentially affect fine sediment distribution of a nearby offshore sand shoal also

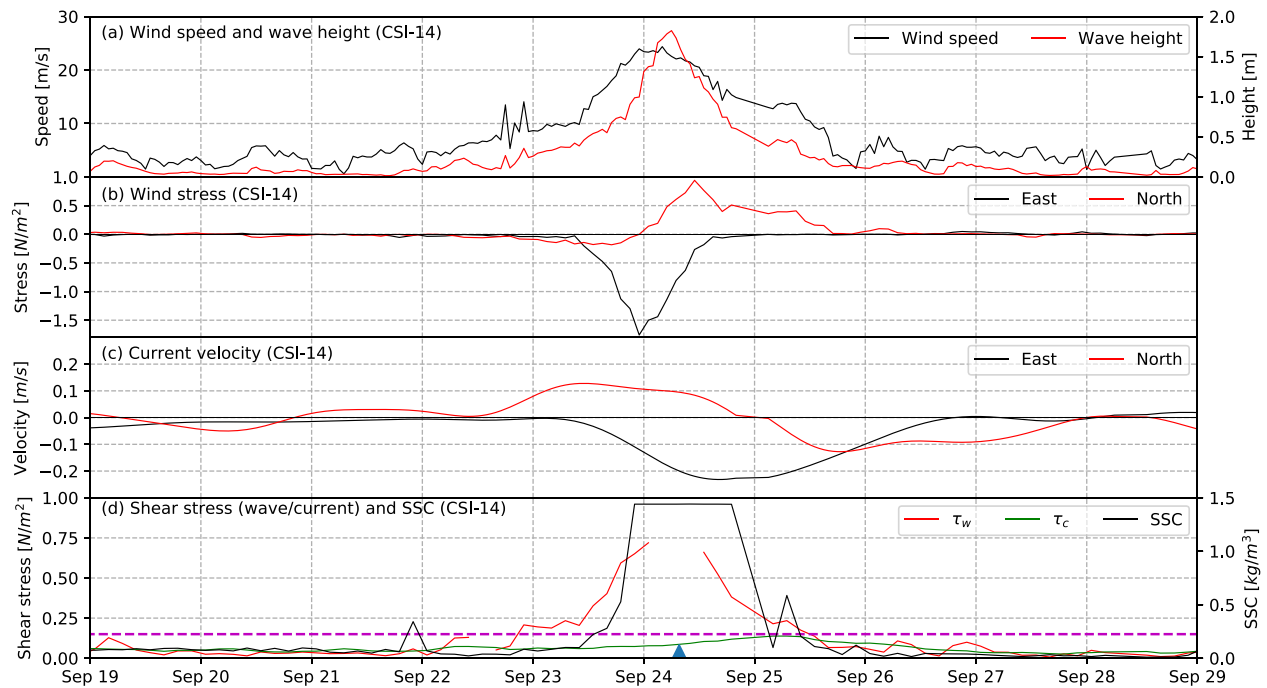


Fig. 8. A time series of (a) wind speed and wave height, (b) wind stress, (c) current velocities, (d) shear stress due to waves (τ_w) and current (τ_c) as well as SSC at CSI-14 for the period between September 19 and September 29, 2005. Purple dashed line in (d) represents threshold for sediment suspension (i.e. critical shear stress). Blue triangle indicates the time when Hurricane Rita made landfall.

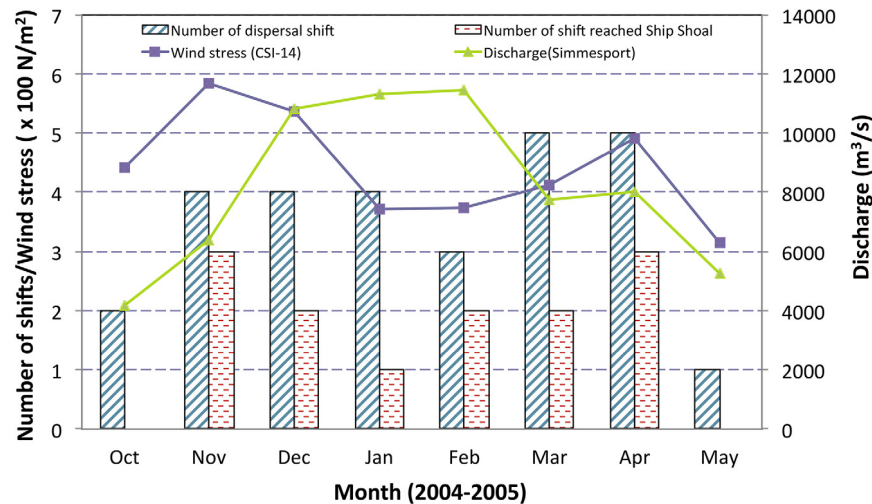


Fig. 9. Monthly mean wind stress at CSI-14, monthly mean streamflow at Simmesport, the number of dispersal shifts, and the number of the shifts that reached Ship Shoal.

known as Ship Shoal. The study region, a broad, low-energy shelf, is characterized by two representative storms and high sediment load from the nearby river. Since the storms drive highly variable winds depending on their tracks and orientations (Fig. 3), the study region provided a unique opportunity to investigate the subject.

Two key processes influencing the dispersal patterns were identified as sediment suspension and advection due to storm winds that progressively change their direction as they move. While the prevailing direction of fine sediment transport along the Louisiana coast is west (i.e. downcoast. Wells and Kemp, 1981), it was revealed that these variable storm winds play an important role in the re-distribution of fine sediment on the shelf.

For the winter storms, the advection due to the strong post-frontal winds ($\tau_{wind} > 0.2 N m^{-2}$) distributed fine sediment from

the bay to the shelf. Sediment suspension prevailed during the post-frontal phase (Fig. 3). Thus, the source of fine sediment should be both the sediment from the Atchafalaya River and re-suspended sediment. In spring when high river discharge coincides with the passage of cold fronts across the region, the fine sediment can be *effectively* transported to the shelf from the bay.

On the other hand, during the period when tropical cyclones approached the coast or made landfall in the vicinity of the bay, the dispersal patterns depended on the intensity of the storms and their tracks. The common feature of the sediment dispersal for both types of hurricane is that both accompany strong sediment suspension and offshore sediment transport due to hurricane-induced waves and advection. They re-distribute the sediment on the bay/shelf, though both have different mechanisms for the offshore transport. During the summer hurricane

season when river discharge is generally the lowest of the year (Fig. 2), it was found that local sediment re-suspension plays an important role in sediment dispersal by supplying fine sediment to be redistributed and it occurs only during high-energy events such as tropical cyclones as discrete events.

How much fine sediment is originated from sediment suspension from bay bed for the dispersal during a storm compared to the sediment load from the river is unknown. We made a rough estimate for a winter storm, which passed on March 29, as an example. The maximum SSC of CSI-14 during the storm was approximately 0.4 kg m^{-3} (Fig. 6). Let us assume that strong suspension occurred and the re-suspended sediment was mixed uniformly throughout the water column in the entire bay, which has an area of approximately 2000 km^2 and the average depth of 3 m. This should be a reasonable assumption for tropical cyclones (Sheremet et al., 2005) and also for strong winter storms (Allison et al., 2000) (Fig. 5). The estimated suspended sediment mass would be 2.4 million tons. The sediment mass from the river for summer months (June–August) estimated from Fig. 2 is roughly 8.0 million tons. Thus, the mass from the re-suspended sediment during a single storm event accounts for more than a quarter of the total sediment load from the river for the entire summer. The percentage becomes even higher for tropical cyclones (e.g. 82.5 percent for H. Katrina and 108.8 percent for H. Rita). These numbers suggest that sediment re-suspension during storms potentially provides a significant amount of fine sediment to be available for transport. Given that the width of the bay entrance is approximately 45 km, the amount of sediment transported to the shelf from the bay during a single winter-storm event while the offshore advection ($\sim 0.15 \text{ m s}^{-1}$) prevails (a couple of days according to Fig. 6), would be approximately 1.4 million tons. The result indicates that more than half of the re-suspended sediment can be potentially re-distributed to the shelf due to the offshore advection during a winter storm. The results highlight an important aspect of sediment dispersal driven by sediment suspension and advection on the bay and shelf.

In-situ data indicated that re-suspended sediment reached surface in 5 m of water; while, the re-suspension was limited to about 5–10 m above the bottom in 20 m of water on the inner shelf during cold front events (Fig. 5). Applying a simple model of wind-induced currents developed by Csanady (1982), the estimated turbulent Ekman layer, $d (=0.1u_*f^{-1})$ where u_* is wind shear velocity ($=1.4 \text{ cm s}^{-1}$) and f is the Coriolis parameter ($=7.29 \times 10^{-5} \text{ s}^{-1}$), is 19.2 m. Given the exponential decay of the current profile, the current flux of the top 10 m accounts for approximately three quarters of the total flux within the layer. Thus, it is suggested that only sediment suspension on the shallower part of the shelf ($<10 \text{ m}$) in tandem with storm-induced current should significantly contribute to the sediment dispersal during the winter storms. The in-situ data showed that surface and bottom cross-shore currents were in phase at CSI-3, but were often out of phase at CSI-6 during winter storms, supporting the above suggestion (data not shown). The region coincides with the area where a large quantity of fine sediment is thought to be deposited as identified by Xu et al. (2011); therefore, the dispersal associated with the recurring sediment suspension during storms is expected. On the other hand, sediment suspension from the deeper part of the shelf is expected to contribute to sediment dispersal only during severe storms such as tropical cyclones.

The aforementioned dispersal patterns, as corroborated by the satellite images (Fig. 3), indicated an implication of sediment dispersal for the exposure of fine sediment on nearby sand shoals. As suggested by the result of the repeated sediment samples from Ship Shoal (Kobashi, 2009), the period of exposure and influence of fine sediment on the shoal environment should be short and ephemeral (only days to weeks). It is thought that

the frequent passage of cold fronts promotes re-suspension and transport of fine sediment freshly deposited on the shallow shoal before they are settled permanently. The relationship between wind stress, river discharge and the frequency of the sediment that reached the shoal suggests an important contribution of post-frontal winds to the fine sediment supply to the shoal. The findings from this study can apply to other sand shoals on the Louisiana shelf such as Tiger and Trinity shoals. Those shoals are also reported to be ecologically sensitive (Stone et al., 2009) and the dispersal patterns are believed to play a role in the benthic habitats of the shoals since some benthic organisms favor sandy bottom over muddy bottom and fine sediment reduces light penetration to the bottom affecting photosynthesis of benthic microalgae that provides dissolved oxygen, which may be a key on the survival of benthos during summer hypoxic periods (Stone et al., 2009).

6. Conclusions

Storm-driven dispersal of fluvial fine sediment and its potential link with sedimentary environment of an offshore sand shoal was investigated based on extended *in-situ* met-ocean data and high-resolution satellite imagery for the period October 2004 to September 2005. Wind stress during both storms had a profound effect on controlling the sediment dispersal patterns. Regardless of the type of storms, common features of the storm-driven sediment dispersal were sediment suspension and offshore advection. The offshore advection in combination with sediment suspension and sediment load from the river effectively distributed fine sediment from the bay to the shelf. In addition, sediment suspension and transport were prevalent on the shelf during tropical cyclones, albeit in an intensified manner, and it further helped re-distribute fine sediment on the shelf. The contribution of the re-suspended sediment as a sediment source for the dispersal has been identified. The data presented in this paper elucidate fine sediment transport associated with variable storm winds and high sediment load from the river over a low-energy shelf, which has not been well documented.

The repeated sediment sampling from the shoal confirmed fine sediment accumulation on Ship Shoal in the previous study. The dispersal mechanism qualitatively revealed a possible pathway of fluvially derived fine sediment to the shoal, which has long been recognized as a sandy shoal. The findings provide valuable insight on how variable winds and morpho-hydrodynamic processes influence sediment dispersal of the bay and shelf ecosystems such as those on the offshore sand shoals. The shoals are known to be viable sand resources to restore rapidly deteriorating barrier islands and are also reported to have a unique benthic habitat.

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